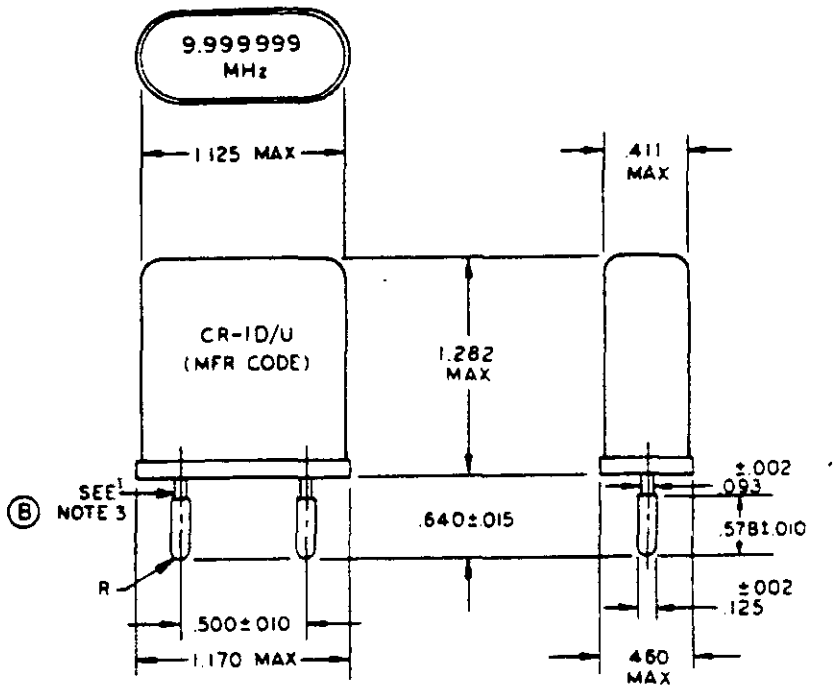
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-1D/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

Pertinent characteristics. 3.6 to 10 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.500	12.70
.010	.25	.578	14.68
.015	.38	.640	16.26
.093	2.36	1.125	28.58
.125	3.18	1.170	29.72
.411	10.44	1.282	32.56
.460	11.68		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
Ⓑ 3. The pin undercut may be omitted.

Ⓑ denotes change

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-11A/U (MIL-H-10056/17)
Dimensions, marking, and configuration	See figure
Frequency range	3.6 to 10 MHz, inclusive
Frequency tolerance (operating temperature range)	±50 PPM
Equivalent resistance	See table III
Antiresonance, load capacitance	35.0 ± 0.5 pF
Mode of oscillation	Fundamental
Operating temperature range	-55° to +90°C, inclusive
Test set, calibration values, and rated drive level	See table II
Shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	± 10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	± 10 percent
Temperature run:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent

1/ Method 201 of MIL-STD-202.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values		Rated drive level
		Resistance	Crystal current	
	MHz	Ohms	mA	mW
TS-330/TSM	3.6 to 5.1	80	25	50 ± 5
	5.1+ to 7.5	48	25	30 ± 3
	7.5+ to 10.0	25	35	30 ± 3

TABLE III. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
3.6 to 5.5	90
5.5+ to 6.5	60
6.5+ to 7.8	50
7.8+ to 10.0	30

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NOTE: This crystal-unit type is to be used for special applications and for replacement purposes only. Specific approval of the Government is required prior to each intended use. See MIL-STD-683.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SG, SM
Navy - EC
Air Force - 11, 85
DSA - ES

User activities:
Army -
Navy - AS, MC, SH
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-0408-39)

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